

18-646: How to Write Fast Code II

Project 2:- Project Midterm Report

The course project is meant for students to have practice in using the GPU on a practical problem of their choice. Specifically, students are required to analyze their application, identify how the problem can be best mapped to the architecture, and implement their design.

Expected Project Outcome

The expected output of the course project is a fast and parallel implementation of the chosen problem that has been *designed* based on available hardware features, and the impact of the hardware features on performance. A fast implementation where implementation choices cannot be explained **will not** meet the requirements of this project. Similarly, a faster implementation obtained by switching to a different algorithm is also not acceptable.

Instructions. For this project, the second deliverable is the formation of the project team, and proposal for the project. A PDF three page document with the following requirements is required to be submitted:

1. *Performance Baseline.* Provide baseline performance plots for the algorithm(s) you have identified. This is to ensure that you have working code that you can compare against. *As such, you should not be writing this code from scratch.* . The data set use to create these plots should also be the same data set used to test your implementation.
 - (a) Provide performance numbers for different input data types or sizes if applicable.
 - (b) It is acceptable if the baseline performance is based on a sequential piece of code running on the CPU or GPU, or even running in Python.
 - (c) If you have 3 different kernels/algorithms, we expect at least 3 different plots; one for each kernel/algorithm.
2. *Parallelism Analysis and Initial Mapping.*

This is the core requirement for this deliverable. It is an analysis of the parallelism that is available, and how the team wants to exploit it for performance.

 - (a) Describe where are the parallelism that can be exploited in your kernel/algorithm. This description needs to be specific as to how the team intends to exploit the available parallelism to produce fast code on GPU. As part of this description, a discussion on how shared memory is to be used (if required). For example,

We intend to exploit computing these independent data points by mapping them to different threads. Each thread will compute Z data points. The data points are grouped into a region of $X \times Y$ points that is computed with one SM. The intermediate data points are stored in shared memory in column-major order.
 - (b) *Provide reasons why you chose to parallelize them this way. If a reimplementaion of data structures or data layout is required, please describe the layout and the rationale for then here as well..*
 - (c) For each kernel/algorithm, please specify who is responsible for the analysis of individual kernel. If multiple people contributed to the analysis, provide an estimate of the split in terms of effort (in percentage).

Submission. As this is a group project, it is acceptable and expected that everyone contributes to the writing of the proposal. However, the same proposal must be submitted *individually*.